VISHWAKARMA
AWARDS
A Maker Bhavan Foundation Initiative

MAKER
BHAVAN
FOUNDATION

VISHWAKARMA AWARDS 2026-27

VK26-IITH-SM-0275

Problem Context

GUIDELINE SLIDE

Project Overview:

Title: PRAN RAKSHAK

Team ID: **VK26-IITH-SM-0275**

Theme 5: Sustainable Mining: Worker Health, Safety & Environmental Hazard Tracking

Team Name: Codemium

Team Leader: Kaaviya Varshini V

Problem Statement: Mine workers face undetected gas and dust exposure underground, and mines discharge toxic runoff that contaminates surrounding water, because monitoring is manual, fragmented across separate systems, and unable to locate a worker when something goes wrong. We need one low-cost, offline capable system that senses gases, dust, and runoff quality, locates workers underground, and warns before limits are breached.

Problem Context

Core Innovation:

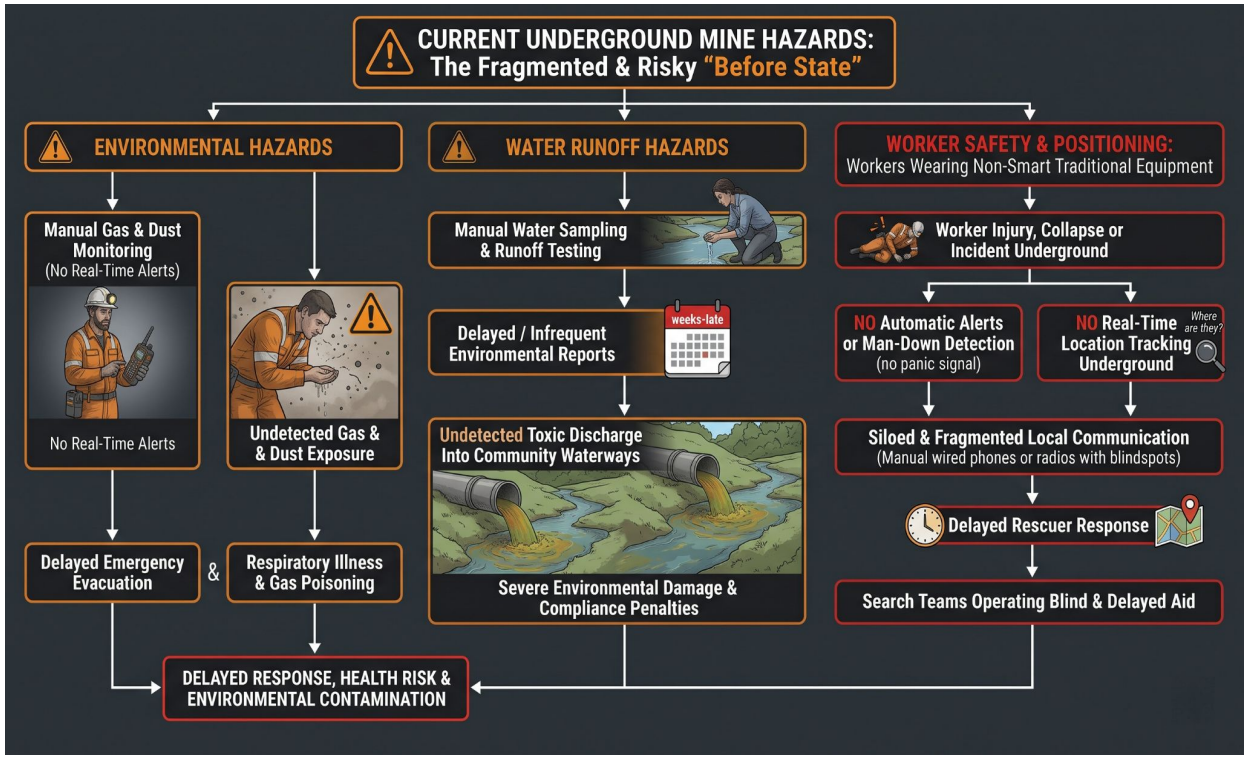
An affordable, offline capable smart safety helmet that integrates real time gas & dust hazard alerts, worker localization, panic button with man down detection & direct wireless warnings to protect miners underground.



User Journey

GUIDELINE SLIDE

Visual "Before State" Requirement



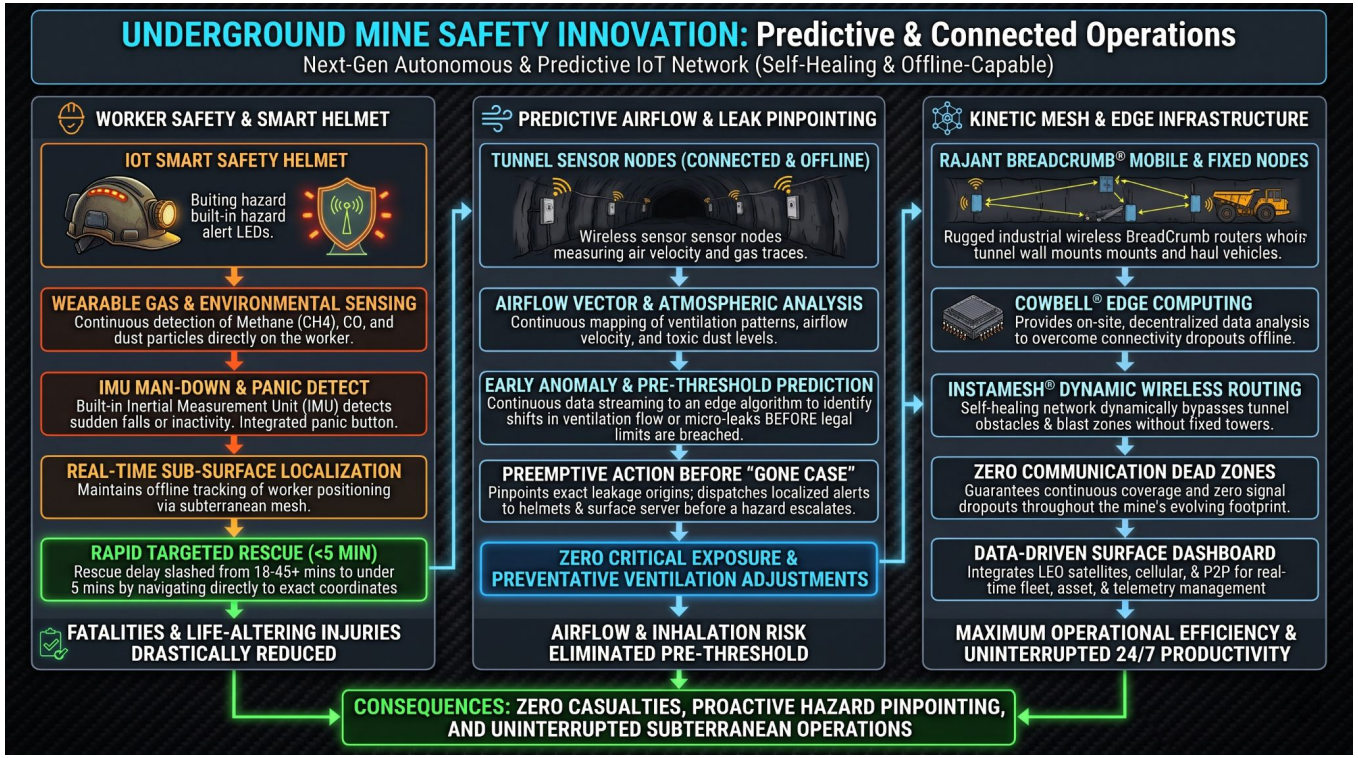
Target Beneficiary & Before-State User Flow

- **Target User: Underground mine workers** (face miners, loaders, drillers), **technical crews**, and **emergency rescue squads** in subterranean galleries and shafts.
- **Operational Bottleneck: Manual, fragmented hazard monitoring** leaves toxic gas and dust unnoticed, while **communication dropouts** leave missing workers unlocated during emergencies.
- **Proposed Future-State Flow:** Nodes and smart helmets stream data -> **Edge models detect airflow shifts and trace gases** to **pinpoint leaks pre-threshold** -> **Direct hazard alerts** flash on helmets -> **Falls or panic triggers** broadcast sub-surface coordinates -> Rescuers reach workers in **< 5 minutes**.
- **Sector Constraints & Capabilities:** Bypasses GPS/LTE dropouts using an **offline, self-healing kinetic mesh network**, enabling **real-time localization** and **gas monitoring** without surface connectivity.

User Journey

GUIDELINE SLIDE

Visual "After State" Requirement



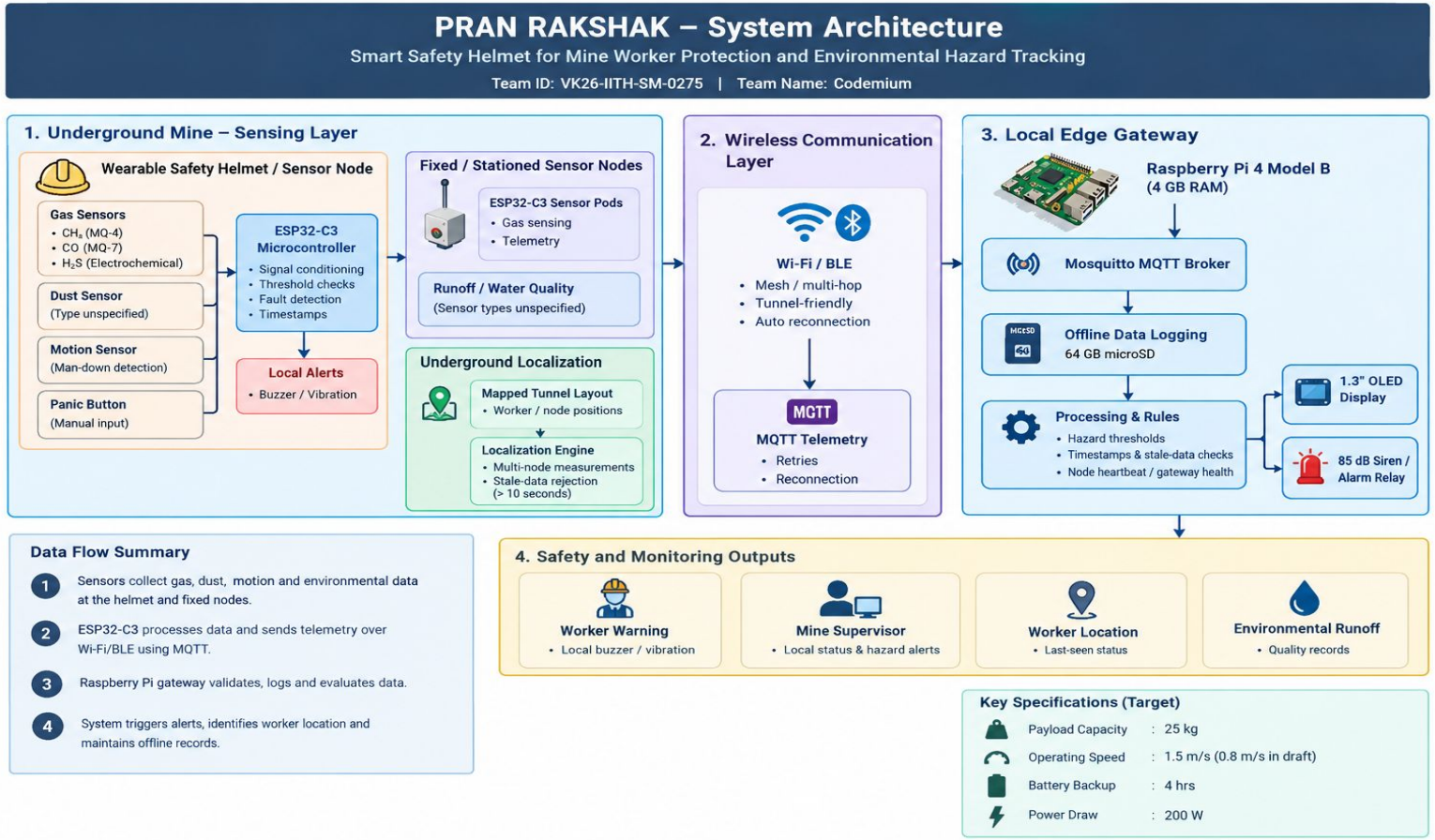
Target Beneficiary & Proposed Future User Flow

- **Target User: Subterranean mine workers** (drillers, loaders, face miners), underground crews, and **emergency rescue squads**.
- **Operational Bottleneck: Undetected toxic gas/dust** from fragmented monitoring, causing **blind search delays** during communication dropouts.
- **Proposed Future Flow:** Sensor nodes & helmets stream air data -> Edge models detect airflow shifts to **pinpoint leaks pre-threshold** -> **Direct visual/audio warnings** flash on helmets -> **IMU fall/panic triggers** transmit coordinate ->Rescue teams reach workers in **< 5 minutes**.
- **Constraints & Capabilities:** Bypasses GPS/cellular dead zones using an **offline, self-healing mesh**, delivering uninterrupted hazard alerts and tracking with **zero surface connectivity required**.

System Architecture & Specs

GUIDELINE SLIDE

System Block Diagram



Target Specifications

PRAN RAKSHAK targets under 10 s end to end gas alert latency (**threshold crossing to helmet vibration and OLED warning**) at 1 Hz sampling of **CH4, CO and H2S**. Each wearable node is designed for 8 hours of battery backup (one full shift) on a **3.7 V 2500 mAh LiPo**, using duty-cycled sensor heaters to keep average draw near 1 W. The IP67-targeted pod weighs about 120 g (52 × 88 × 28 mm). **Nodes report to a Raspberry Pi 4 gateway over Wi-Fi/MQTT using kinetic mesh**, and the gateway runs fully offline with at least 7 days of **local logging**. A lost node is **flagged within 30s**, man-down triggers after 30 s of no motion, and **workers are localized to the nearest node zone.**

Hardware Selection



Deploy Sensor Nodes

- Wearable sensor node integrated into the worker's helmet
- Fixed sensor nodes installed inside the mine
- Wearable node contains MQ-4 methane, MQ-7 carbon monoxide, electrochemical H_2S, dust, motion, and panic button sensors
- Fixed nodes monitor CH_4, CO, H_2S, runoff, and water quality

Collect and Process Data

- ESP32-C3 microcontroller for data collection, signal conditioning, threshold checking, fault detection, timestamping, and telemetry

Transmit Wireless Data

- Communicate through Wi-Fi and Bluetooth Low Energy (BLE)
- Use mesh or multi-hop communication to transmit readings and alerts

Manage Edge Gateway

- Raspberry Pi 4 Model B with 4 GB RAM and Mosquitto MQTT broker
- 64 GB microSD card for offline data logging
- Process data using hazard threshold detection, timestamp validation, stale-data rejection, and health monitoring

Output Safety Alerts

- 1.3-inch OLED display for real-time status and alerts
- 85 dB siren for audible hazard warnings
- Mine supervisor interface for notifications
- Worker localization module using tunnel map

Example: System Architecture & Specs

PROJECT TERRAWEED EXAMPLE



Engineering Target Specifications & Subsystems

Parameter	Target Spec	Selected Hardware / Protocol
Gas sensing	CH4, CO, H2S at 1 Hz sampling	MQ-4 (CH4), MQ-7 (CO), H2S electrochemical sensor, analog conditioning into the ESP32-C3 ADC
Alert latency	<10 s, threshold crossing to helmet alert	On-node threshold check; helmet vibration motor and OLED
Node controller	160 MHz, deep sleep between samples	ESP32-C3-WROOM-02-N4 (RISC-V, Wi-Fi/BLE)
Battery backup	8 hrs (one shift), heaters duty-cycled	3.7 V 2500 mAh LiPo, TP4056 charge and protection, low-dropout 3.3 V regulator
Link and protocol	Lost node flagged in 30 s, auto reconnect	Wi-Fi + MQTT (QoS 1), topics, `sensors/<node_id>/telemetry`, `/status`, `gateway/alerts`
Gateway	Fully offline, 7+ days local logging	Raspberry Pi 4 (4 GB), Mosquitto broker, threshold rules, local dashboard, 64 GB microSD

Architecture Rationale: ESP32-C3 with Wi-Fi and MQTT was chosen for low-cost, offline-capable communication between sensor nodes and the Raspberry Pi gateway, enabling real-time hazard alerts and worker monitoring.

Node & Gateway Architecture Specs

GUIDELINE SLIDE

- **ESP32-C3 Sensor Nodes:** Wearable & stationed pods with ESP32-C3 (RISC-V, Wi-Fi/BLE) for low-power telemetry.
- **Hazardous Gas Sensing:** Integrate calibrated CH4, CO, and H2S sensors with intrinsic safety and analog preconditioning.
- **Raspberry Pi Server:** Deploy local Raspberry Pi base station for offline data logging, MQTT brokering, and alerts.



Example: ESP32-C3 Node Hardware & Enclosure

PROJECT TERRAWEED EXAMPLE



Figure: CAD assembly of wearable ESP32-C3 sensor pod with gas diffusion manifold & IP67 seals

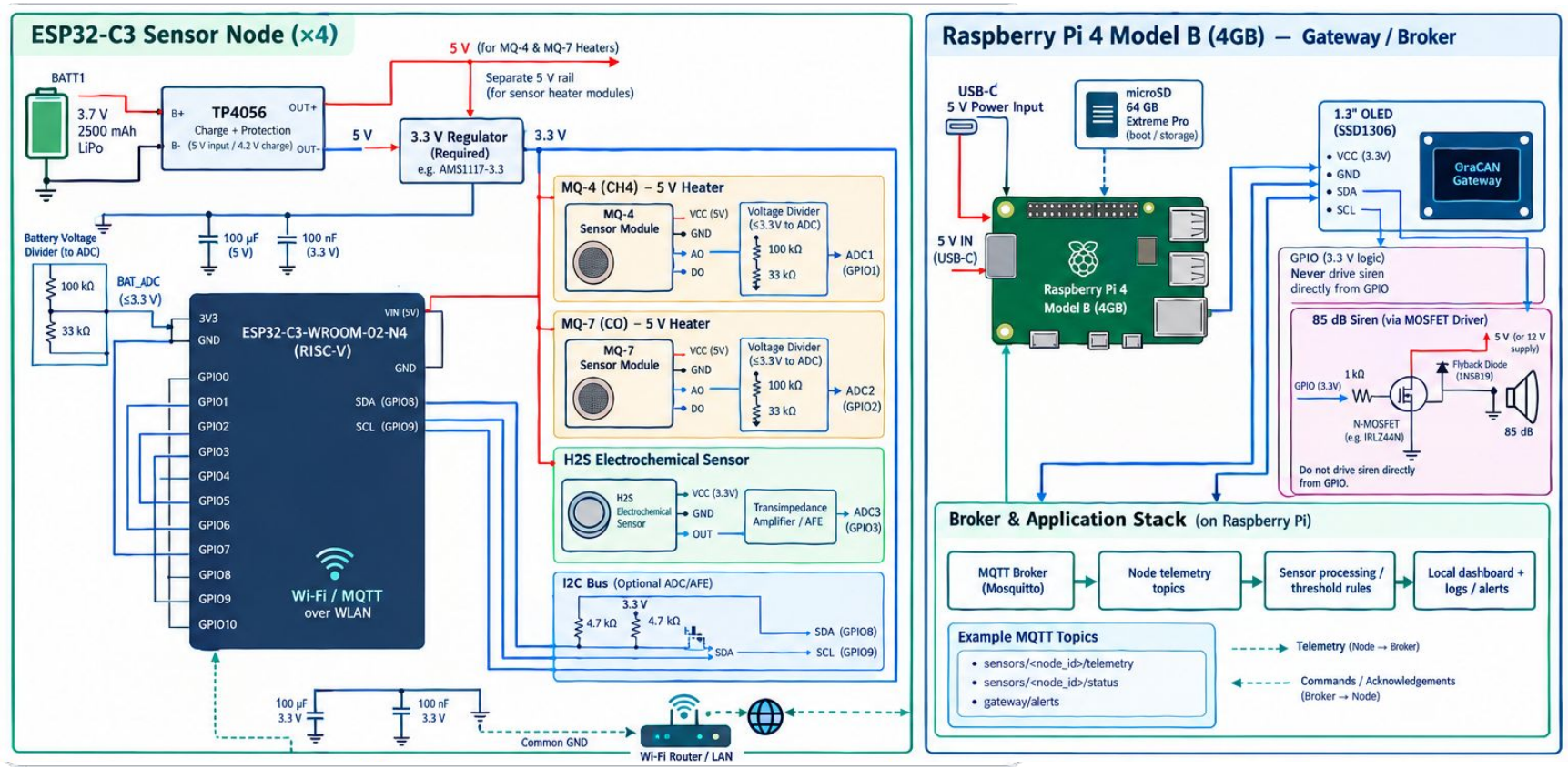
Hardware Specifications & Enclosure Design

- **Enclosure Material:** Polycarbonate-ABS blend (UL94 V-0 flame-retardant) with silicone gasket; IP67 ingress rated against toxic slurry & mine dust.
- **Core Processing Unit:** ESP32-C3 RISC-V 32-bit MCU @ 160MHz; hardware crypto accelerator, low deep-sleep draw, and native 2.4 GHz Wi-Fi/BLE 5 mesh.
- **Gas Intake & Filtration:** Sintered brass/stainless steel flame arrestor mesh over MQ-4 (CH4), MQ-7 (CO), and H2S chambers preventing thermal spark ignition.
- **Power Architecture:** 3.7V 2500mAh Li-Po battery with intrinsic safety fuse, current limiting, and low-quiescent LDO; achieves >8 hours per shift.
- **Impact Integrity:** Drop-tested for 2.0m shock impacts onto granite; shock-absorbing silicone dampeners isolate internal sensor PCB and antenna.

Enclosure Optimization: Vortex gas diffusion port reduces response latency from 8.2s to 2.1s while deflecting ambient airborne coal particulates.

Electronics Schematics & BOM Details

GUIDELINE SLIDE



Circuit Schematics: ESP32-C3 pinouts, ADC gas sensor conditioning, 3.3V LDO regulation, and TP4056 charging.

Gateway Architecture: Detailed Raspberry Pi server wiring, MQTT broker network interfaces, alarm relays, and 5V/3A supply.

Itemized BOM Table: Listed all parts, exact MPNs, quantities per node, unit costs, and domestic vendor sourcing lead times.

Electronics Schematics & BOM

PROJECT TERRAWEED EXAMPLE

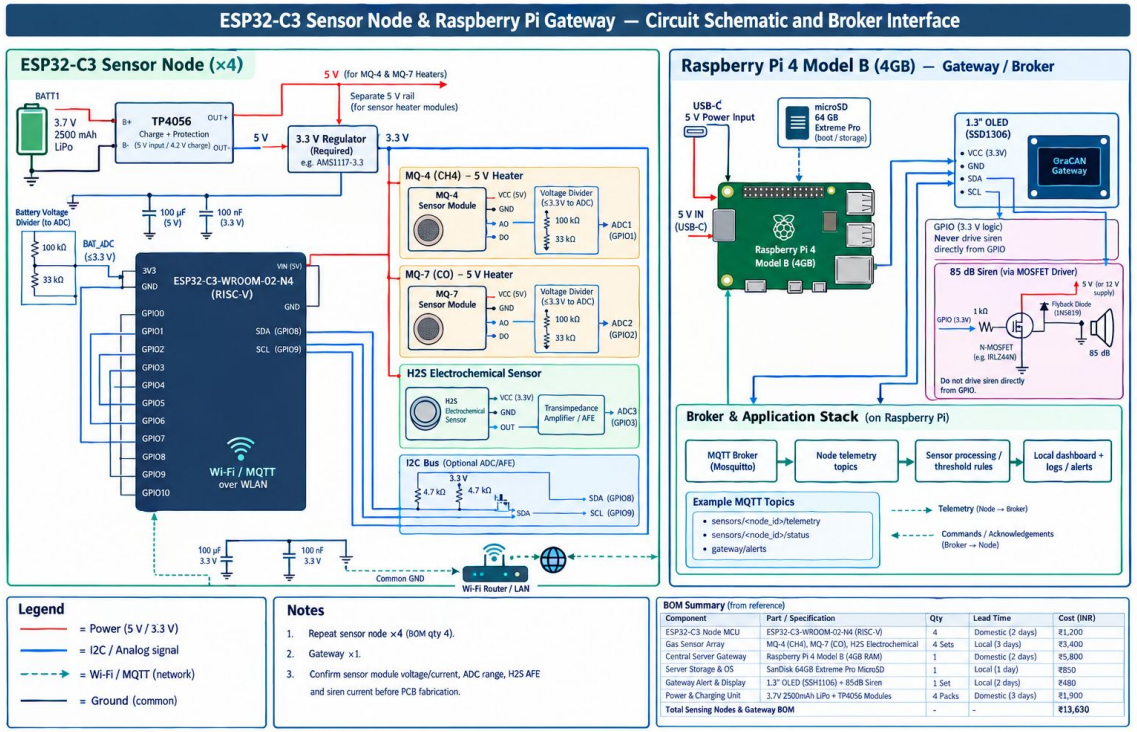


Figure: Circuit schematic of ESP32-C3 sensor node & Raspberry Pi gateway broker interface

Itemized Prototype Bill of Materials (BOM)

Component	Part / Specification	Qty	Lead Time	Cost (INR)
ESP32-C3 Node MCU	ESP32-C3-WROOM-02-N4 (RISC-V)	4	Domestic (2 days)	₹1,200
Gas Sensor Array	MQ-4 (CH4), MQ-7 (CO), H2S Electrochemical	4 Sets	Local (3 days)	₹3,400
Central Server Gateway	Raspberry Pi 4 Model B (4GB RAM)	1	Domestic (2 days)	₹5,800
Server Storage & OS	SanDisk 64GB Extreme Pro MicroSD	1	Local (1 day)	₹850
Gateway Alert & Display	1.3" OLED (SSH1106) + 85dB Siren	1 Set	Local (2 days)	₹480
Power & Charging Unit	3.7V 2500mAh LiPo + TP4056 Modules	4 Packs	Domestic (3 days)	₹1,900
Total Sensing Nodes & Gateway BOM	**₹13,630**	**-**	**-**	**-**

Procurement Status: All ESP32-C3 modules, gas sensors, and Raspberry Pi gateway hardware sourced domestically with standard lead time under 7 days.

Your Maker Journey

GUIDELINE SLIDE

Authenticity Verification



Raw Repository Linkage

Google Drive / GitHub repo link:

It includes files of KiCAD Schematics

https://drive.google.com/drive/folders/15X_x7dBfrwkh-Dxh-tOKbknpaBlmvr9N?usp=sharing

It includes KiCAD PCB
https://drive.google.com/drive/folders/14xDrxjfyS91qGC_mshUwT2NkXfo6L4GV?usp=sharing

It includes AutoCAD
https://drive.google.com/drive/folders/1SFh4BCX0wZwIrOmNC6qaMer0btrjspzQ?usp=sharing

GitHub Repo link
https://github.com/SrivarsanK/PranRakshak

FEA Metrics:

file:///C:/Users/HEMA/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/0F95DE038A8191096EFC20E4BF04F771E63E5B
transfers/2026-43/Report.html

PRAN RAKSHAK / MECHANICAL STUDY

Enclosure lid: mesh refinement and response

Preliminary, assumption-based analysis. Actual STEP geometry and numerical solves; material, loading and rigid supports are illustrative. No material-strength or safety pass is claimed.

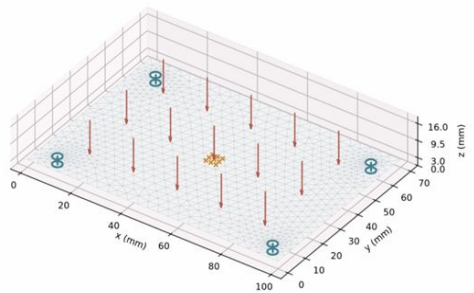
Centre deflection	Final tetrahedra	Selected mesh criterion
0.06623 mm	**7,411**	**Met**
Under 10 N assumed distributed load	42,564 displacement DOFs	Two successive comparisons below 2%

1. Geometry and study setup

Named lid extracted from VKA2026.step: 100 × 70 × 3 mm, four 3.5 mm holes. Isotropic material E=2000 MPa, ν=0.35, both assumed. All displacement components fixed on the four screw-hole walls. Uniform downward top-face traction totals 10 N.

The teal marks identify fixed screw-hole supports. Red arrows indicate downward loading. Amber marks the plan locations of nine stress probes inside the upper portion of the lid.

Actual lid geometry and assumed loading/supports



Teal: fixed screw-hole walls. Red: uniform downward (-z) traction; arrow length is schematic.
Total force = 10 N; E = 2000 MPa; nu = 0.35, all assumed.
Amber marks the plan locations of fixed stress probes inside the upper portion of the lid.

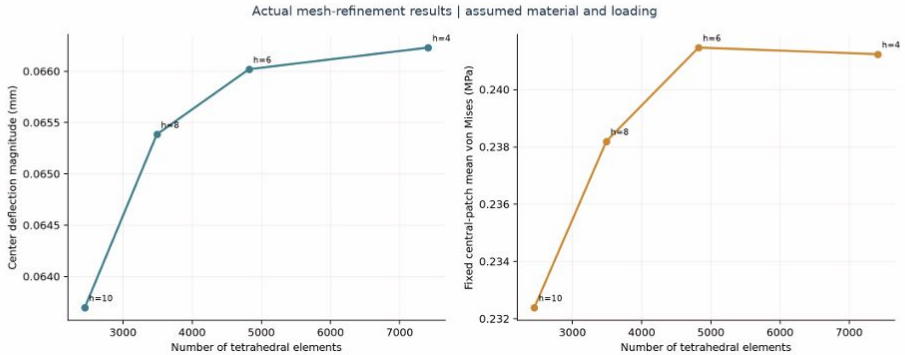
2. Mesh statistics

Target h (mm)	Tetrahedra	Geometric vertices	P2 field nodes	DOFs	Centre abs(uz) (mm)	Probe mean VM (MPa)	Deflection change	Probe stress change
10	2,449	772	4,630	13,890	0.0636966	0.232387	-	-
8	3,490	1,062	6,473	19,419	0.0653845	0.238192	2.581%	2.437%
6	4,822	1,508	9,116	27,348	0.0660195	0.241465	0.962%	1.355%
4	7,411	2,353	14,188	42,564	0.0662305	0.241233	0.319%	0.096%

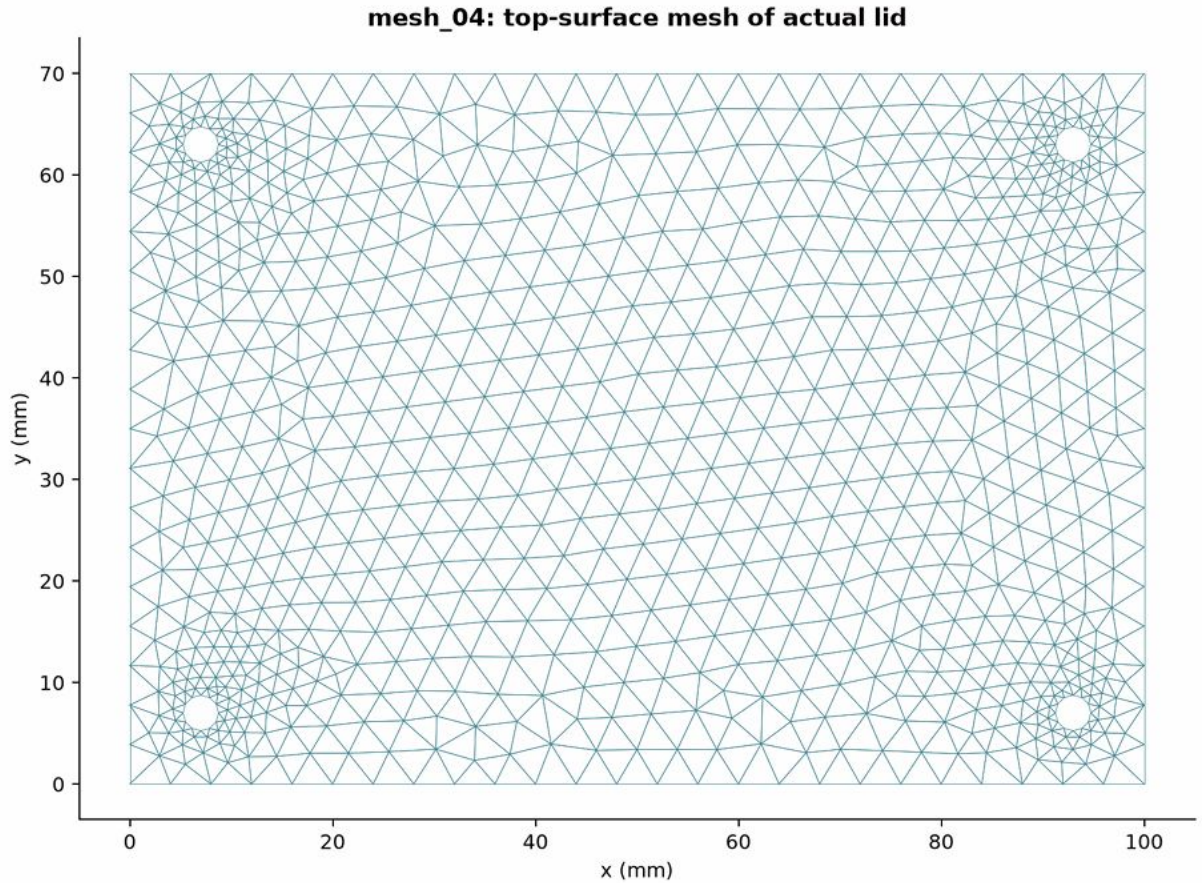
Gmsh stores linear geometric vertices; the solver adds quadratic displacement edge nodes. h is the upper mesh-size control. Curved walls are faceted. Full statistics: CSV.

3. Selected response convergence

Centre displacement and mean von Mises stress at nine fixed central points, z=2.7 mm. These measures changed by 0.319% and 0.096% in the final comparison. This is a successive-mesh criterion, not an absolute error bound.



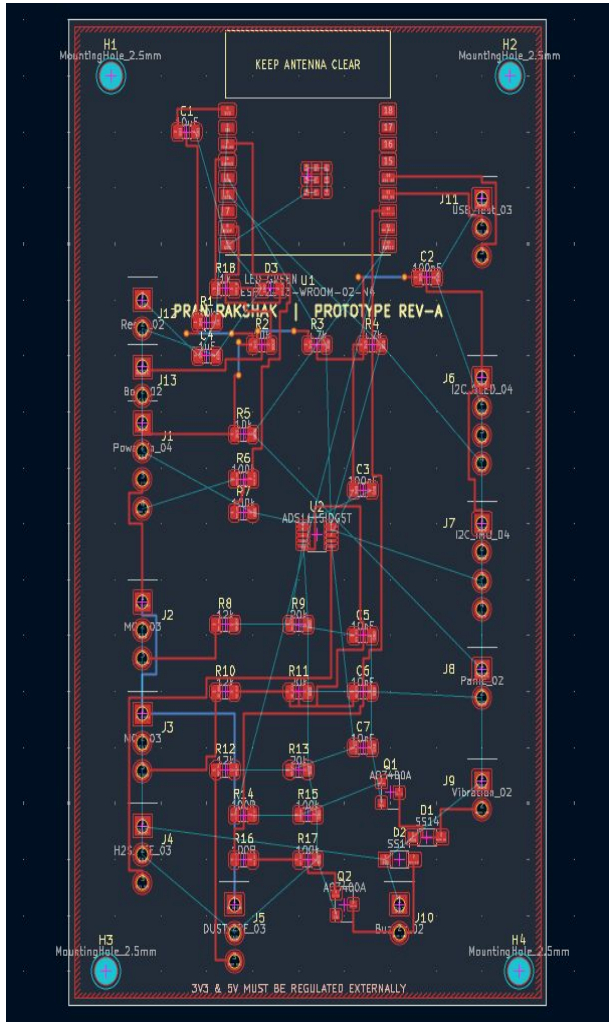
4. Final mesh

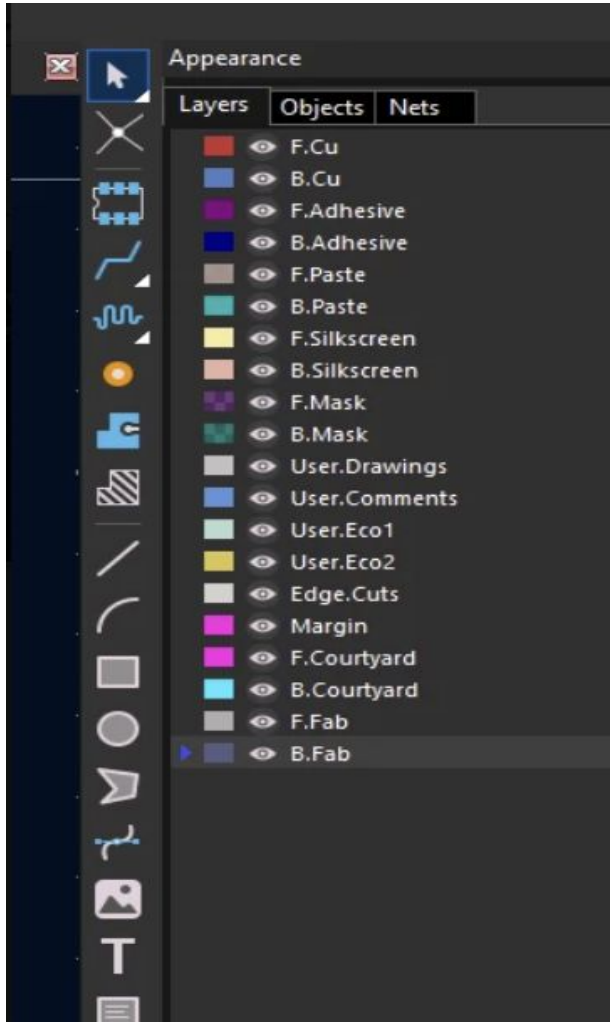


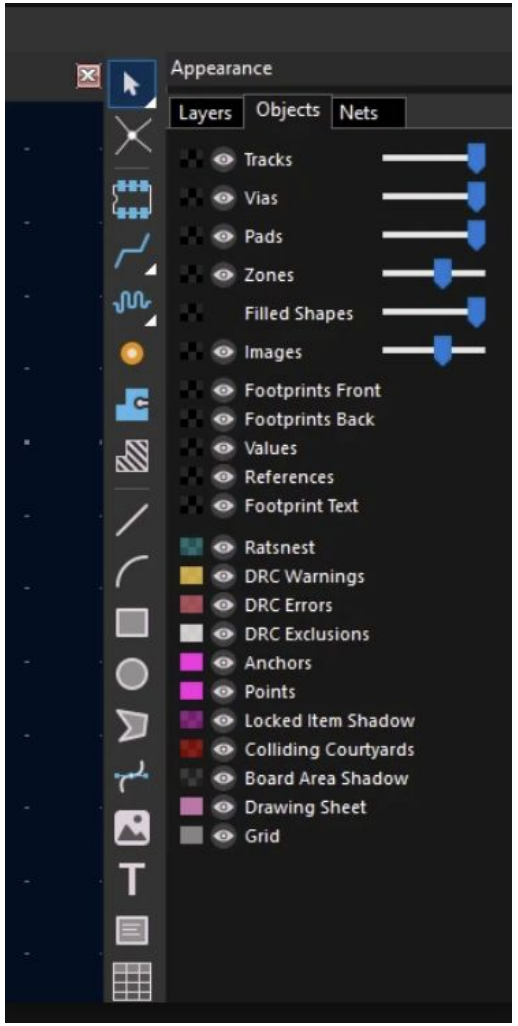
7,411 tetrahedra | 2,353 geometric vertices | 14,188 quadratic displacement nodes
Minimum SICN: 0.338 | Straight-sided geometric cells; quadratic displacement field

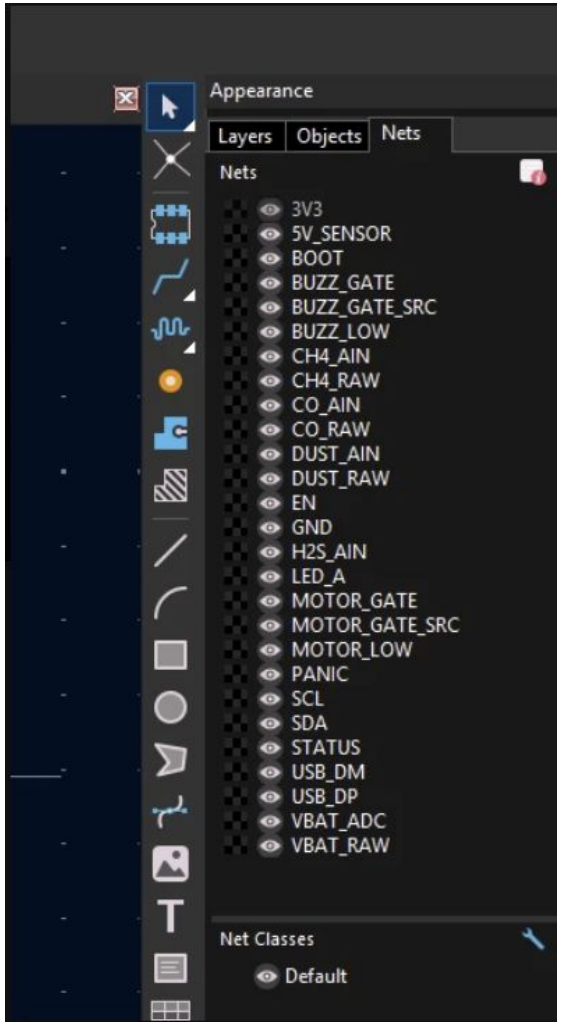
PCB KiCad:

https://drive.google.com/drive/folders/14xDrxjfyS91qGC_mshUwT2NkXfo6L4GV?usp=sharing





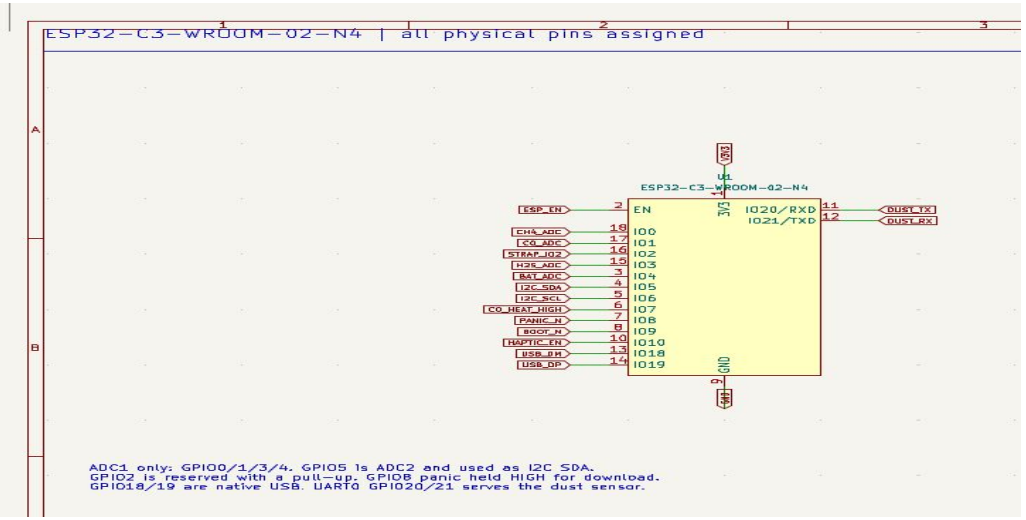




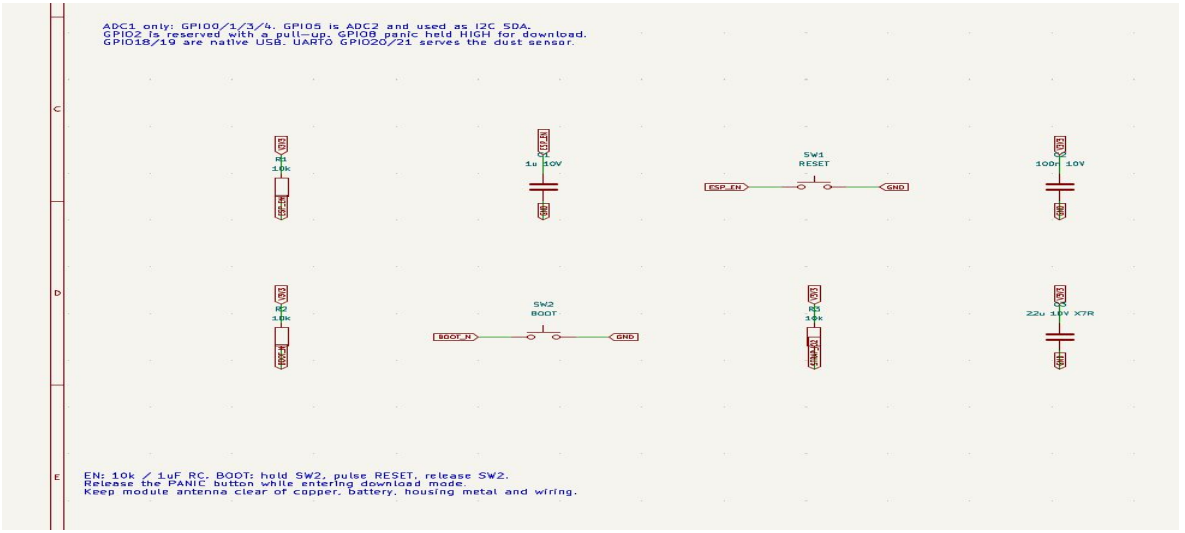
Work-In-Progress Software Evidence:
https://drive.google.com/drive/folders/15X_x7dBfrwkh-Dxh-tOKbknpaBlmvr9N?usp=sharing

Screenshots of original design assets:

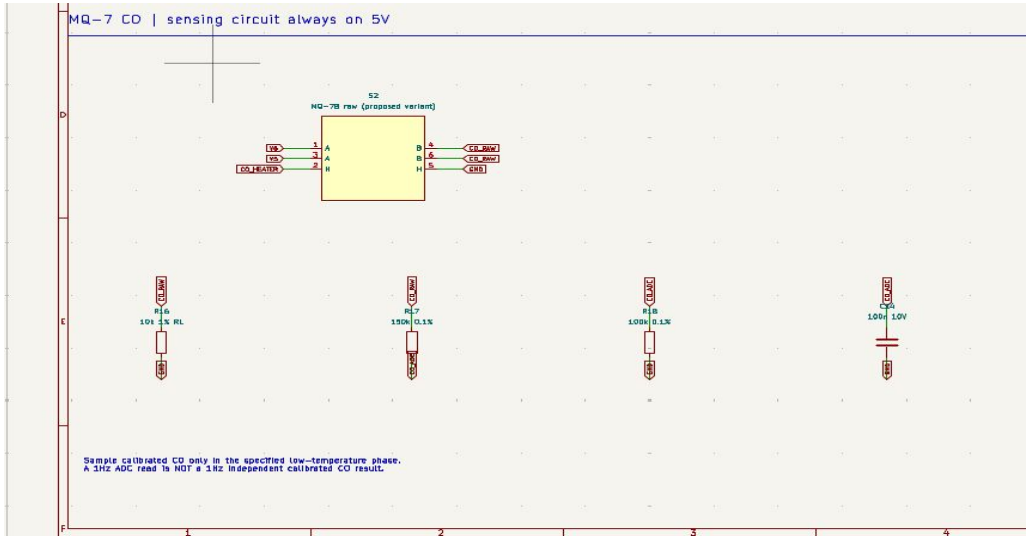
Overview



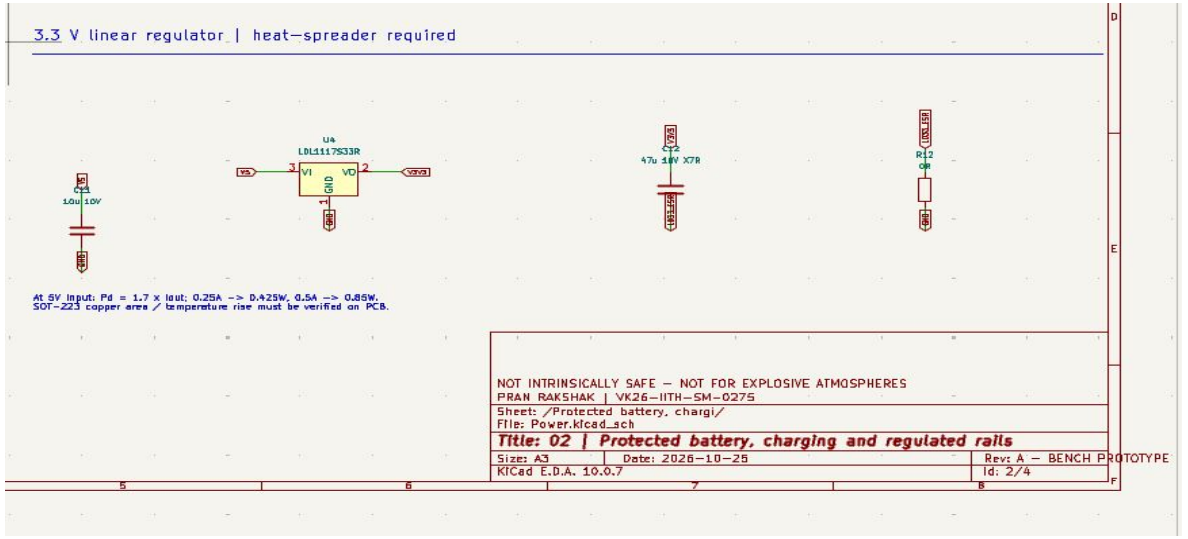
Protected Battery charging and regulated rails



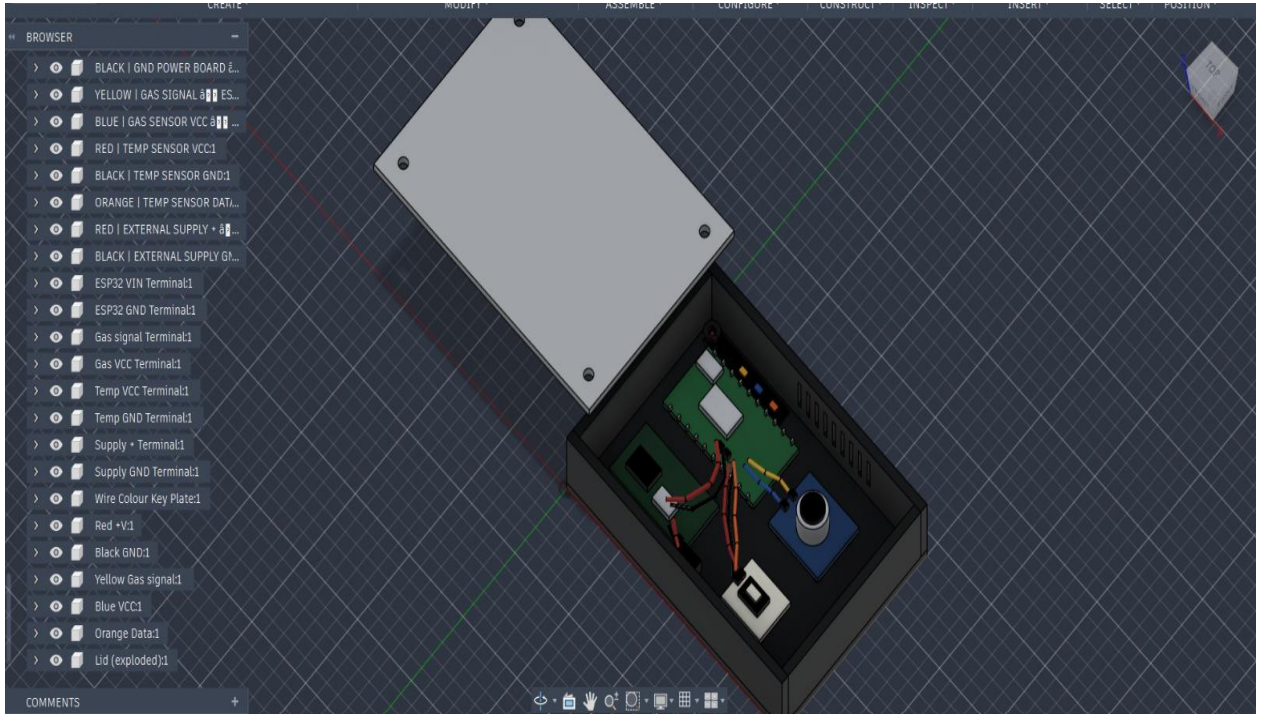
Gas sensor Conditioning and MQ-7 heater cycle

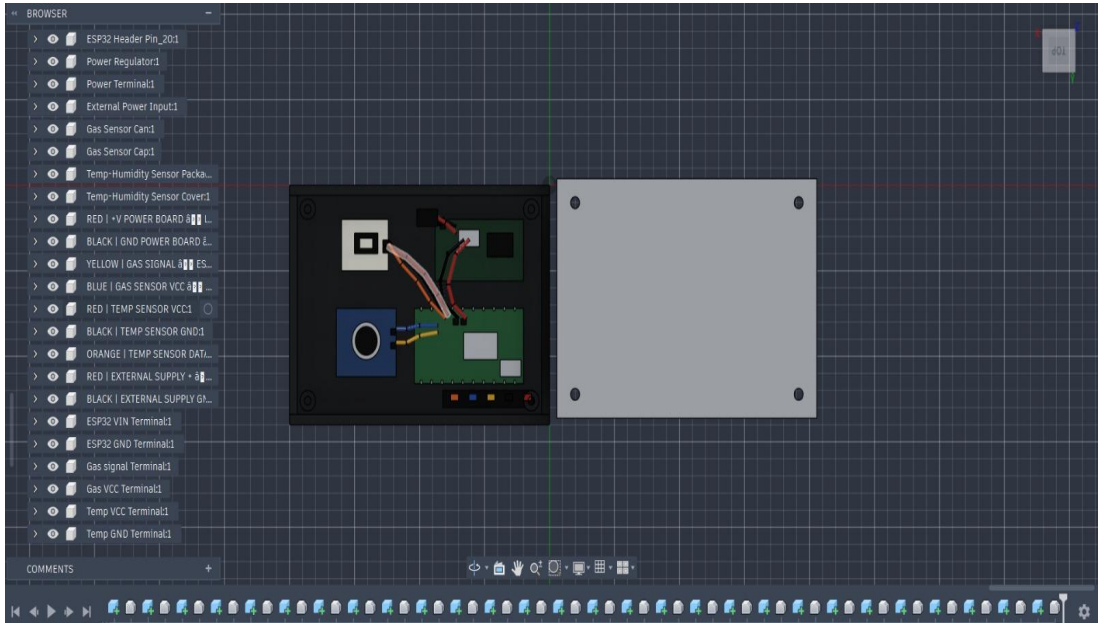


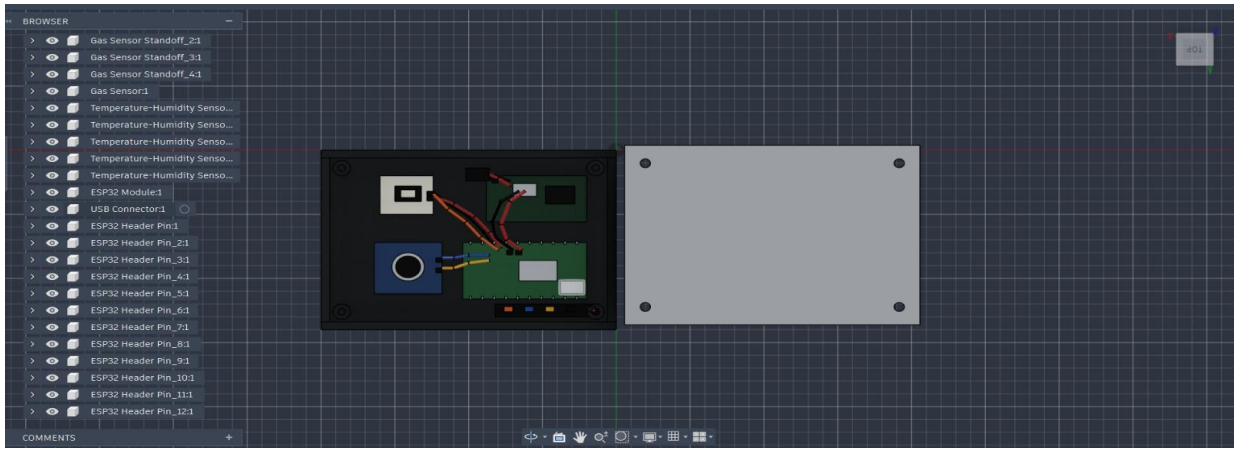
Dust, Motion, Display, Panic and haptic alert

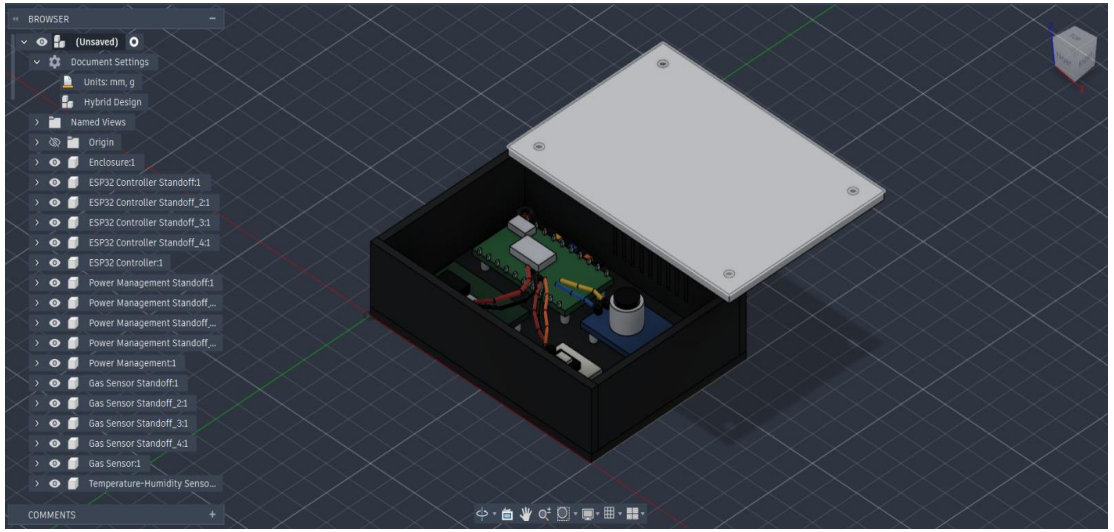


AutoCad:https://drive.google.com/drive/folders/1SFh4BCX0wZwIrOmNC6qaMer0btrjspzQ?usp=sharing









Maker Journey

PROJECT TERRAWEED EXAMPLE



Stage 1 to Stage 2 Engineering Iterations

- **Design Evolution:** Developed the PRAN RAKSHAK concept into an offline-capable mining safety system integrating gas sensing, worker monitoring, emergency alerts, and underground localization.
- **Hardware Milestone:** Selected the ESP32-C3 sensor nodes, MQ-4 for methane, MQ-7 for carbon monoxide, an H_2S sensor, and a Raspberry Pi 4 gateway for local monitoring and data logging.
- **System Integration:** Designed the sensor-node electronics, power regulation, battery charging, gas-sensor conditioning, and MQTT-based communication architecture.
- **Safety & Reliability:** Incorporated local hazard alerts, sensor-fault detection, low-battery warnings, automatic reconnection, and offline data buffering to improve reliability in underground environments.
- **CAD Milestone:** Developed detailed electronic schematics and hardware enclosure designs for the ESP32-C3 sensor nodes, incorporating gas-sensing chambers, protected battery charging, voltage regulation,dust-resistant sealing.

Verified Drive Repository:

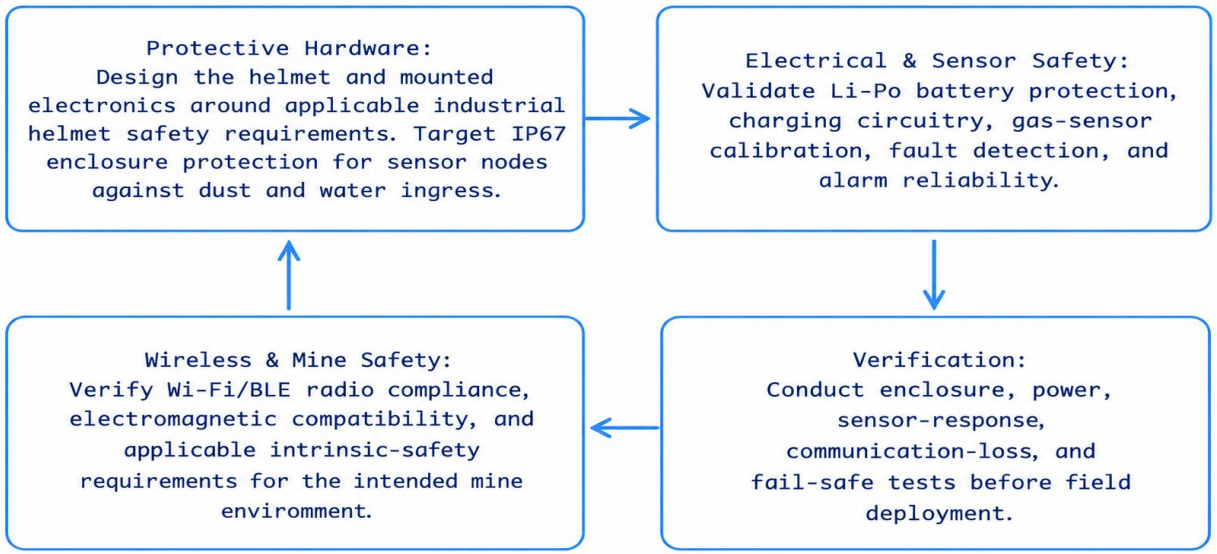
https://drive.google.com/drive/folders/15X_x7dBfrwkh-Dxh-tOKbknpaBlmvr9N?usp=sharing

(Contains KiCad v8 schematics, and PCB KiCad).

Institutional Sign-Off: Work audited and certified by Faculty Prof. Rebita John, Electrical & Electronics Eng. of SRM IST University.

Safety Mitigation

Standards Compliance



Potential failure mode	Risk / effect	Embedded software protection
Gas sensor drift or failure	Incorrect hazard readings	Calibration checks, threshold validation, sensor fault alerts
Power drop or low battery	Sensor or node shutdown	Battery monitoring, low-power warning, controlled shutdown
Wireless communication loss	Delayed hazard alerts	Heartbeat monitoring, automatic reconnection, local alarms
Sensor data delay	Late hazard detection	Timestamp validation, stale-data detection, watchdog timers
False or missed alarms	Unsafe worker response	Threshold validation, filtering, alarm self-tests
Gateway/server failure	Loss of central monitoring	Local node level alerts, offline data logging, gateway health monitoring

Hazard Mitigation



Risk analysis

Assess gas-sensor faults, missed alerts, power loss, communication failure and false alarms using the nodes based on the ventilation blueprint.

Fail-safe response

Keep local alarms active and log critical events without internet using slave/master controllers supporting emergency response.

Failure detection

Monitor sensor health, low battery and disconnected nodes; flag unreliable readings.

Instant local alerts

Trigger on helmet's OLED and vibration alerts when validated threshold limits are reached to the master controller.

Integrated Ventilation Safety Solution

Ventilation safety:
Combining airflow monitoring with wireless connectivity for rapid alerts & data driven ventilation management.

Airflow detection:
Real time airflow sensing identifies changing ventilation conditions, enabling faster detection of inadequate airflow and supporting proactive mine safety

Real-time data processing:
Cowbell edge computing supports local data processing for responsive monitoring applications.

Kinetic Mesh Connectivity:
Multi-radio, multi-path wireless networking maintains communication between mobiles, environmental sensors & monitoring systems across the mine.

Continuous coverage:
Nodes & mesh routing adapt to changing site conditions, helping maintain connectivity as equipment & operations move.

Integrated safety alerts:
Connecting airflow sensors to the mesh network enables timely transmission of abnormal condition alerts to operators & control rooms.

Made with Napkin

Roadmap

Design FMEA (Failure Mode and Effects Analysis)

Failure Mode	Potential Hazard	Embedded Fail-Safe / Mitigation Response
Tiller Stone JamGas sensor drift or dead sensor (MQ-4 / MQ-7 / H2S)	False “safe” reading, so a worker stays in a methane/CO/H2S zone	Baseline-drift check at every boot; reading stuck or flat-lined for >60 s flags the sensor faulty; node raises a “sensor fault” alert and never shows “safe”
Node loses link to gateway (tunnel bends/branches)	Worker unlocatable, alerts not delivered	Local threshold check on the ESP32-C3 with buzzer/vibration alert, no server needed; auto-reconnect plus MQTT retries; gateway flags the node “lost” if last-seen >30 s
Stale or clock-skewed readings	Wrong localization, wrong leak source	Every packet timestamped; gateway discards readings older than 10 s before running localization

Fabrication Phase (Months 1 - 2)

Oct 2026: Finalize the schematic and run the KiCad DRC. Order the PCB and components. Bench-calibrate the MQ-4/MQ-7/H2S sensors against a reference gas.

Nov 2026: Assemble nodes and helmet enclosure. Build the Pi 4 gateway, the Mosquitto broker and the threshold rules. Bring up the OLED and siren.

Integration & Field Demo (Months 3 - 4)

Dec 2026: Test mesh and link-loss behavior. Tune man-down detection. Run the watchdog and brownout tests. Run a first full multi-node localization test on a mapped corridor.

Jan 2027: Run a long-duration soak test (battery life, drift). Final tuning and a rehearsed live demo for Stage 3.

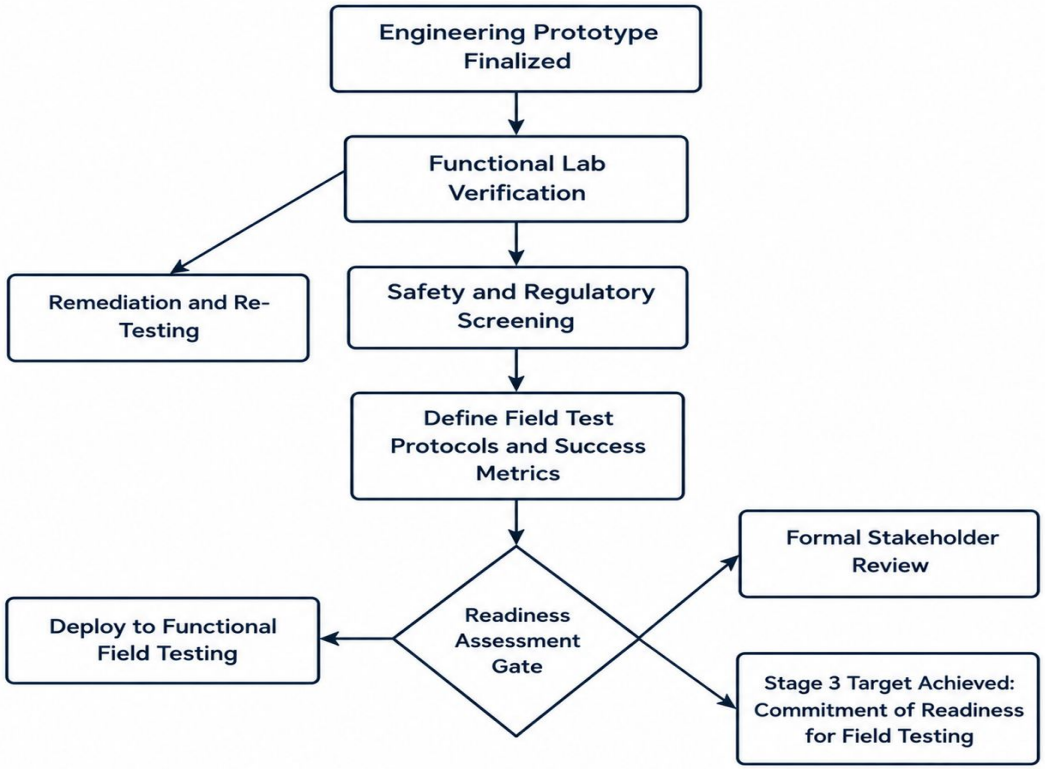
Roadmap

GUIDELINE SLIDE

4-Month Fabrication Plan

Fabrication Phase (Months 1-2):
The first two months deliver a working hardware. We finalize the schematic, **run the KiCad DRC** and **order the PCB**, then **calibrate the MQ-4, MQ-7 and H2S sensors** against a reference gas so every reading has a known error. We then **assemble four sensor nodes, the helmet unit and the Raspberry Pi 4 gateway**, and **bring up the MQTT broker, threshold rules, OLED**. Exit criterion is **every node sends timestamped telemetry to the gateway**, and a gas threshold crossing **triggers a local alarm with even minimal internet.**

Integration & Field Demo (Months 3-4):
The last two months turn the working hardware into a system. We **test mesh behavior** and **kinetic mesh connectivity**, and run the watchdog and brownout tests. We then **run a multi node localization test on a mapped corridor using the ventilation blueprint** and a long duration test for battery life and sensor drift. Stage 3 in which the **gas alert, panic button, man alert and node loss flagging all work as per the stated targets.**



Technical Challenges & Risks

Sensor accuracy and Calibration

Sensor can be affected by the surrounding environment and can produce less accurate outputs and respond slowly

Poor **placement of the sensors** may cause issues and cause detection problem

Airflow Direction and Velocity

As we are using the direction of air to find the source of leak, there can be many situations where the flow changes and we predict wrong.

Synchronization/Server delay

Nodes may have different clock offsets, while gas sensors can take time to respond to concentration changes.

Network Reliability

The network may be interrupted continuously due to bends and branches of caves.

Power Limitations

ESP32-C3 memory, GPIO and ADC limitations; sensor power requirements; unstable supplies; and potential overheating or resets.

Potential Solutions & Risk Mitigation

GUIDELINE SLIDE

Potential Solutions

- **Sensor Accuracy & Drift Control:** Uses **multi-point calibration**, **noise filtering**, and **temperature-humidity compensation** to eliminate false readings.
- **Optimized Spatial Coverage:** Combines **3D-mapped fixed nodes** at calculated intervals with **smart helmets** for continuous mobile monitoring.
- **Predictive Pre-Threshold Pinpointing:** Analyzes **airflow velocity and directional vectors** to **pinpoint leak origins** before statutory limits are reached.
- **Autonomous Kinetic Mesh:** Deploys **self-healing BreadCrumb nodes** to route around subterranean obstacles with **zero fixed-tower dependency**.
- **Offline Edge Computing:** Executes models directly on **on-site edge hardware**, delivering **zero-latency alerts** without surface connectivity.
- **Hardware & Timing Reliability:** Prevents sync drift via **capture-timestamping**, while **hardware watchdogs** eliminate brownouts and memory crashes.

Risk Mitigation

- **Multi-Tiered Sensor Fallbacks:** Automatically switches to **catalytic or metal-oxide backup sensors** if primary optical units fail, drift, or saturate.
- **Ventilation Model Fail-Safe:** Reverts the dispersion algorithm to **historical ventilation baselines** to map gas spread if anemometers clog.
- **Physical Component Protection:** Shields sensor modules using **dual-stage sintered metal and hydrophobic PTFE membranes** against dust, slurry, and moisture.
- **Redundant Mesh Transmission:** Fails over to **long-range sub-GHz protocols** to broadcast distress coordinates if high-bandwidth mesh paths are severed.
- **Local Data Buffering:** Caches unsent telemetry in **non-volatile local memory** during disconnections, auto-syncing immediately upon reconnection.
- **Supervisory Hardware & Power Backup:** Employs **external watchdog circuits** to clear system hangs and **battery buffers** to maintain low-power beacon modes during outages.

Technical Challenges & Risks

Sensor Calibration Problems

How it can be solved?
Calibrate each sensors properly
Apply filtering and temperature compensation where supported

Improper placement of Sensors

How it can be solved?
Proper mapping of the tunnel is required before placement
Place sensors at suitable intervals.

Airflow Direction and Velocity

We have to regularly update the localization model when ventilation changes
Measure airflow direction regularly

Server Delay

Synchronize node clocks timestamp every reading.
Detect stale or delayed readings before localization.

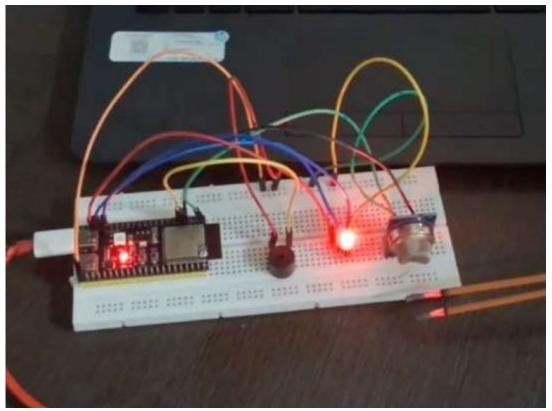
Network Delay

Implement automatic reconnection and message retries.
Monitor node connectivity and last-seen timestamps.

Power Limitations

Add watchdogs, brownout handling and reset diagnostics.
Use a stable, properly rated power supply.

Image Sources



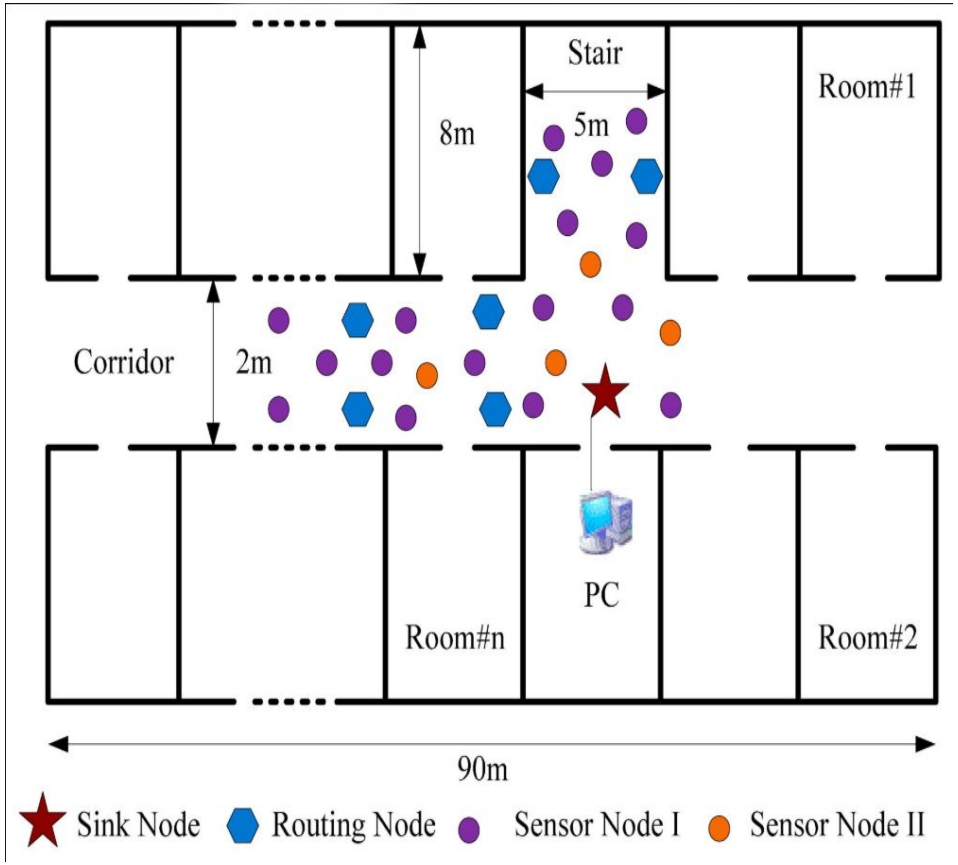
https://www.electronicwings.com/users/KhushaliVadaliya/projects/5953/smart-gas-leak-detector-using-esp32-and-blynk

We can see which sensors are going to be used with our microcontroller.
This can be explained in our system architecture model where we can detect gases using these sensors, the temperature inside and safety of specialists

The diagram shows a rough plan where the sensors are going to be placed in the tunnel.
This is explained in our after state model where our sensors are already placed and result are evaluated
https://pmc.ncbi.nlm.nih.gov/articles/PMC11902780/?



Image Sources



Useful for understanding how different nodes communicate with each other and with the sensors
Explained more about it in our After State model using airflow detection, gas detection and more.
https://pmc.ncbi.nlm.nih.gov/articles/PMC3348832/

References & Sources

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[1] Zhang, Xiang-Dong & Fan, Shi-Xing & Zhang, Hu-Wei & Zhang, Yu. (2013). Mine-Tunnel Design & Monitoring Analysis for Major Faults and Complex Geological Conditions. TELKOMNIKA Indonesian Journal of Electrical Engineering. 11. 10.11591/telkomnika.v11i3.2302. https://www.researchgate.net/publication/270723039

[2] National Institute for Occupational Safety and Health. (2010). Guidelines for the control and monitoring of methane gas on continuous mining operations (DHHS (NIOSH) Publication No. 2010-141). Centers for Disease Control and Prevention.https://www.cdc.gov/niosh/publications/numbered/2010-141.html

[3] National Institute for Occupational Safety and Health. (n.d.). Atmospheric monitoring. Centers for Disease Control and Prevention.https://archive.cdc.gov/www_cdc_gov/niosh/mining/topics/atmosphericmonitoring.html

[4] Pritchard, C., & Shriwas, M. (2020). Ventilation monitoring and control in mines. Mining, Metallurgy & Exploration, 37(4), 1015–1021. https://stacks.cdc.gov/view/cdc/230545

[5] Sadeghi, S., Soltanmohammadlou, N., & Nasirzadeh, F. (2022). Applications of wireless sensor networks to improve occupational safety and health in underground mines. Journal of Safety Research, 83, 8–25. https://doi.org/10.1016/j.jsr.2022.07.016

[6] Damiano, N. W., Homce, G. T., & Jacksha, R. (2014). A review of underground coal mine emergency communications and tracking system installations. National Institute for Occupational Safety and Health / CDC Stacks.https://stacks.cdc.gov/view/cdc/228109

[7] Singh, A., Kumar, D., & Hötzel, J. (2018). IoT-based information and communication system for enhancing underground mines safety and productivity: Genesis, taxonomy and open issues. Ad Hoc Networks, 78, 115–129. https://doi.org/10.1016/j.adhoc.2018.06.008